

RUI QIU

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EDUCATION

Georgetown University

Master of Science in Data Science and Analytics. cGPA: 4.0

Washington D.C., United States

Aug. 2021 – May 2023

The Australian National University

Master of Science in Statistics

Canberra, Australia

Feb. 2017 – Dec. 2018

University of Toronto

Honors B.Sc. with Distinction (Maths, Statistics, Computer Science)

Toronto, Canada

Sept. 2011 – June 2016

SKILLS

Programming Python, R, SQL, Scala, JavaScript, HTML, CSS

Machine Learning XGBoost, scikit-learn, pytorch, langchain, tidyverse, tidymodels, numpy, pandas

Visualization ggplot2, Shiny, d3.js, matplotlib, seaborn, plotly, leaflet

Data Science A/B testing, feature engineering, model interpretability, counterfactual explanation, NLP, time series, hypothesis testing

Big Data & Tooling PySpark, Spark ML, Databricks, MLflow, Delta Lake, AWS (EC2, EMR, S3, SageMaker), Azure, Docker, Linux, Hugo

EXPERIENCE

Navy Federal Credit Union

Principal Data Scientist

Vienna, Virginia

July 2023 – Present

- Built **XGBoost multiclass classification** models for credit card product propensity, deployed end-to-end in **Spark ML pipelines** on Databricks with MLflow tracking and Delta Lake; supports cross-organizational business units with data-driven member targeting.
- Prototyped LLM-based pipelines for call center content processing within the data science department, evaluating summarization and intent classification capabilities for operational efficiency.
- Expanded the MLOps engineering toolkit with reusable utility functions and documented standards, reducing onboarding friction and promoting consistent data science practices across teams.
- Developed model interpretability artifacts — including **SHAP** analysis and **DiCE** counterfactual explanations — to satisfy Model Risk Management requirements and translate complex model behavior into actionable narratives for non-technical stakeholders.

International Finance Corporation (World Bank Group)

Data Scientist

Washington, D.C.

May 2022 – May 2023

- Leveraged GPT-3.5 and **LangChain** to enable contextual Q&A on IFC's ESG reports, saving ~90% of analyst extraction time.
- Fine-tuned **ClimateBERT** on climate-related downstream tasks, improving fact-checking accuracy by 6%.
- Integrated the model into MALENA (Machine Learning ESG Analyst), demonstrating real-world utility for NLP tasks.
- Designed interactive dashboards with stakeholders for clearer insight communication.

Georgetown University

Teaching Assistant / Bootcamp Manager

Washington, D.C.

Aug. 2021 – Aug. 2022

- Graduate TA for ANLY-503 (Advanced Data Visualization) and ANLY-560 (Time Series); held weekly office hours and graded for 30+ students.
- Facilitated course materials, managed the course Slack channel, and organized the course GitHub organization.

K2L Canberra

Product Data Analyst

Canberra, Australia

July 2019 – June 2021

- Used Python and SQL to track product data, increasing recommendation accuracy by 15%.
- Built MySQL/R visualizations for KPIs, cutting manual reporting time by 3 hours per week.
- Improved UX metrics by 20% via increased feedback collection and rapid iteration on customer needs.
- Launched 6 client products (websites, mobile and web apps) in 2 years, contributing to a 65% revenue increase.

K2L

Co-founder / Data Analyst

Canberra, Australia

Feb. 2019 – June 2019

- Led a team of 12 in developing an online Q&A community blending StackOverflow and TikTok concepts; grew to 1,000 users in 3 months.
- Conducted market competition analysis using scraped data to inform initial product planning.
- Developed functional prototypes that improved client communication efficiency by 30%.

PROJECTS

yasqat — A Python package for sequence analytics, supporting multi-task analysis of behavioral event streams with a modular, composable API. [[Repository](#)]

What Ingredients Are You Tasting In Authentic Japanese Flavors? — Comprehensive interactive visualization on Japanese cuisine with D3, plotly and leaflet. [[Website](#)]

Basketball Analytics and Beyond — A project on various basketball-related topics, both on-court and off-court. [[Website](#)]

tarantino — An R package generating color palettes inspired by Quentin Tarantino's films for visualization. [[R-Weekly](#)] [[Repository](#)]